3rd ESA Security Conference

14-15 APRIL, 2025

KEY TAKEAWAYS

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ESPI European Space Policy Institute

POLSA

Ministry of Economic Development and Technology
Republic of Poland

About the 3rd ESA Security Conference

The **3rd ESA Security Conference**, held under the auspices of the Polish Presidency of the Council of the EU, was co-organised by the Polish Ministry of Economic Development and Technology, the Polish Space Agency, the European Space Agency, and the European Space Policy Institute, on the 14th & 15th of April 2025 in Warsaw.

This 3rd edition—unlike the first and second, which focused on user needs—was designed as a **political conference**, coinciding with a critical shift in Europe's policy following the Russian aggression in Ukraine, and echoed by the Polish Presidency's rallying call "**Security, Europe!**". Nowhere is this broader shift clearer than in the European space sector, where security and defence were long overlooked and underfunded by most EU countries and other main European Space actors, despite a growing dependence on space. The speakers underscored the urgent need to reverse this trend in the face of a war on European soil and insecure relations with the United States.

To do so, **concerted and joint efforts are paramount**; leveraging Member States, the EU, ESA, and industry to accelerate and federate capability development. This requires establishing **interoperability** between national and European initiatives, overcoming fragmentation, enhancing **operational readiness** to meet both civil security and defence needs, and establishing a 5th freedom to harness the full potential of Europe as an **innovation powerhouse**. With **over 380 participants**, significant media presence and over 30 high-level speakers, including active and former Prime Ministers, Ministers, Members of Parliament, Secretaries of State, to C-suite Industry Executives and Heads of Space Agencies, **ESPI with ESA identified the following seven messages** that stand out as the conference's defining takeaways:



7 KEY TAKEAWAYS

1 Business as Usual is Over

In 2022, the Eastern border of Europe became a battlefield, prompting nations to strengthen their resilience and sovereignty with the mission to restore peace to the continent. By 2025, the very foundations of the European security architecture are called into question, with transatlantic relations facing their greatest strain ever. Amidst this shift, the European space sector must react swiftly to develop the acutely needed autonomous space capabilities for Europe's security and defence needs, towards more independence. This must happen in the shortest time possible, and by embedding innovation and emerging technologies at the core to futureproof Europe's future security and defence architecture and of course to plan for the longer term.



2 Innovation at the Core

To secure a strategic advantage, Europe must treat innovation not as a by-product, but as a driver of both its security and prosperity. While scaling existing capabilities, the continent must significantly scale up and accelerate its R&D efforts to stay ahead across a portfolio of rapidly evolving technologies and applications. This requires sustained public support for the full innovation cycle, from R&D to market expansion, through public programmes and by creating market conditions for scaled private investment. Only by thinking bigger and moving faster can Europe turn innovation into a strategic strength, while also conquering new markets globally.



3 It Takes a Village

To successfully tackle this challenge, Europe must collectively leverage the skills and funding depth of all its institutional players and industry, while aiming at increased coherence of action. In this context, new flagship programmes are needed to meet the continent's evolving security and defence needs and to act as a catalyst and aggregator of action, thus minimising market fragmentation as far as possible.

In a European market where countries root for their own national champions, we need to face this reality and leverage both national and collective solutions to form a resilient federated architecture. ESA, in response to requests of its Member States and in synergy with the European Commission, stands ready as an engine of innovation and a prime entity with programme management authority to partner with industry and deliver on public needs on time. The ERS (European Resilience from Space - ESA) proposal towards EOGS (Earth Observation Governmental Service - EU) is a concrete example of this approach.







4 Defence Users as a Catalyst

Ministries of Defence across Europe are increasingly seeking to invest in national space assets and tap into existing commercial or public civil infrastructure. Initiatives like the Atlantic Constellation are already shifting toward defence, revealing an acute need for capacity in the shortest term possible. In this context, it is expected that European security and defence actors will become a key anchor customer for native space solutions, ensuring much-needed market prospects and uptake for both established industry and European startups and scaleups that have emerged over the past years.





5 Dual-Use as a Baseline

In space, the separation between civil and military uses is often more artificial than real as most space systems are inherently dual use, especially at the level of R&D and capability development The distinction is only made at the level of deployment in operational settings or the origin of budgets.

In this context, it is recognised that there is no effective security and defence strategy without space, and any new space programmes need to recognise their role also in a security context. Future policies, procurement schemes, and industrial planning must reflect this reality and overcome institutional and cultural barriers to embrace dual use as a baseline rather than an exception.









6 Space and Defence Industry Partnerships

To meaningfully engage with defence users and fully unlock the dual-use nature of space, companies must, however, navigate a unique and often unforgiving contract regulations imposed by defence customers and defence industry integrators. Yet, defence actors are also seen as more willing to take risks, enabling faster idea-to-market cycles and direct real-world feedback in an environment where strategic edge is paramount. Successful partnerships between space companies and defence industrial actors suggest that innovation can reach security end users more effectively when channelled through established pathways.





7 2025 – A Year to Make a Difference

The urgency to develop and operationalise the full spectrum of autonomous capabilities tailored to Europe's security and defence cannot be overstated, through notably the pooling of capabilities (e.g. a system of systems). In a world moving at unprecedented speed and uncertain direction, any further delay will come at a high cost. 2025 must be the year of bold decisions—before irreversible setbacks take effect.

Space must be recognised as a prerequisite for Europe's future security; decisively supported at the ESA Ministerial Council in Bremen, advanced in the forthcoming proposal of the next EU Multiannual Financial Framework and reflected in strengthened national defence programmes empowered by the escape clause. We have to find the right balance between the legitimate national sovereignty and a collective framework allowing a stronger united Europe.



THANK YOU & SEE YOU IN 2027!

The organisers sincerely thank all participants in the Conference as well as the speakers, who actively contributed to the event by sharing their perspectives, insights and thoughts both on- and off-stage.

We look forward to seeing you at the 4th edition of the conference in 2027!

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Keynote speech of ESPI Director Ludwig Moeller

3rd ESA Security Conference
14. & 15. April 2025
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3rd ESA Security Conference in Warsaw, 14. & 15. April 2025

ESPI Director Ludwig Moeller Keynote Speech 14. April 2025

Welcome to the 3rd ESA Security Conference. It was already pointed out that we are not doing business as usual any longer, as we face a war next door. I would propose that our next conference is in Kyiv. Let's give ourselves the objective of a more peaceful future.

A few words on ESPI, as Europe's independent think tank for space. We also have the privilege of being connected to the reality of our members. We are forward-looking, but at the same time, we are grounded in program realities, as space programs and strategies are being shaped. But now to our conference: I'd like to underline that "security," which is the title of the conference, is much bigger than the military component. Civil protection or civil infrastructure protection is one of many domains where space is needed for security before you would talk about the military domain. And yes, "dual use" is at the core of it all—and I will not talk about dual use, we've done it in the past four years. I think this is understood. It's time to act on it. Acknowledging how close we are to an active war —we are really at a time of substantial seismic change which requires action.

I am just back from the Space Symposium in Colorado Springs — the biggest space event on the planet—driven by, and dominated by, security and defence actors: many US military, but also European and international partners. And the title in fact, "Building Partnership to Secure our Future", was on partnership, and the spirit and the values presented there were on partnership and collaboration. But we do not know yet completely what is about to come, and whether this may be more of a transactional approach to space and policy in general—how this would impact something that is a relationship based on trust over decades between partners.

I want to focus on three things: scale, speed, and village. Village means everybody around the table.

1 VILLAGE

Let me start from the end—from the village. This is about national programs and national funding. It's about ESA, the EU, and industry. 80% of public budgets in European space are in national budgets. 60% of European public funds are implemented through ESA. We have about 15% of private money in the global market—also in Europe. Importantly, we have the operational programs of the EU: for Galileo, for Copernicus, and in the making, IRIS². It is about a federation—a system of systems of all these elements. It is about a coalition of the willing. It's not one design, but it's now a complex interoperation of different components. Inclusivity means we go beyond EU borders—UK, Norway, Switzerland. We have regions like the Arctic Region—Finland, Norway, Sweden, Denmark—coming together to explore solutions.

For industry, civil security and defence are not two different things. And 60% of the security and defence contracts today on the European level are contracted through ESA. Nationally, it's France and the UK in the lead—and that's not a surprise. But we also see, on the industrial side, that some **defence industry actors step up** and come together with space industries. We need to all come together.

2 SCALE

Scale—that's something we try sometimes to put aside. But I quote a former NASA administrator, "Policy is where the budget is". And he also said, "Better a program without a strategy than a strategy without a program." Now—we want both. We do not have a lack of strategies. We may have a lack of programs and scale. If you translate this— we are about at 14 billion Euro of public investment per year in space in Europe. If you compare this to the U.S., we have a factor of roughly five—and probably more. If you compare it to the military sector, the difference is one to fifteen. And the DoD and the SDA budget is now bigger than NASA's budget. Today, we are at 0.07% GDP invested across Europe in space. We at ESPI advocated, two years ago in our **vision ESPI2040**, for public investments to be doubled to 0.15% of GDP. To some it may have looked naive at the time—that this would never happen. In the meantime, we talked to experts—and they ask for 0.15 to 0.25%. Some people call for "whatever it takes". And if you translate this into figures, we're scaling up to about 30 billion a year. And maybe more.

Now—we have positive examples. Germany just established a new ministry. And for the first time in Germany—and maybe in Europe—the name of the ministry, which is Research, Technology, and Space, contains "space" in the name of the ministry. And there is hope. Germany should double its investment in space. And if they did, in GDP terms, Germany would be at par with France.

3 SPEED

The third point: speed. A quote from Colorado Springs: "Policymaking is outpaced by the speed of technological evolution." That is true for AI—it's true for space. So, we're dealing with something very agile. We need to find ways to leverage our agility. Another quote is that "Technology is at the front line of geopolitics and geoeconomics". When it comes to speed, we should all challenge ourselves, it concerns the whole village.

In the next three to five years, we need to make it happen. I really count on the ESA ministerial, what the Commission will be doing on preparation of the MFF, and pre-dating the MFF.

4 PUBLIC-PRIVATE

It is clear-the government's task is to be an anchor customer. Contracts against which investments on the private side can be justified; business plans can be put forward. I would add to this—and it has been said by von der Leyen: the government also needs to be an enabler. We need to make sure that we, as the public side, enable innovation. Without innovation, there will be nothing. We may conceive all the policies we want—obtain all the funding, imagine all the procurements schemes—if our solution is not top-notch innovation, we will not be in the race. We will not be competitive—not on the market and not to secure your peace.

Industry is equally important. Now, industry needs to develop its competitiveness. Industry is confronted with Digital. Space is part of digital. We are increasingly data-centric and information-centric. We need to break the silos between civil and military—also in procurement, also in innovation. Industry should do the scouting out in front of the governments. So go out before the government even understands what they want. It is a risk—but this is the way the system needs to be working. We cannot work only against the user requirements and—because the government requirement will be too slow, and industry will be out of the market if it only relies on government specifications.

5 WHAT DO WE NEED IN CAPABILITIES?

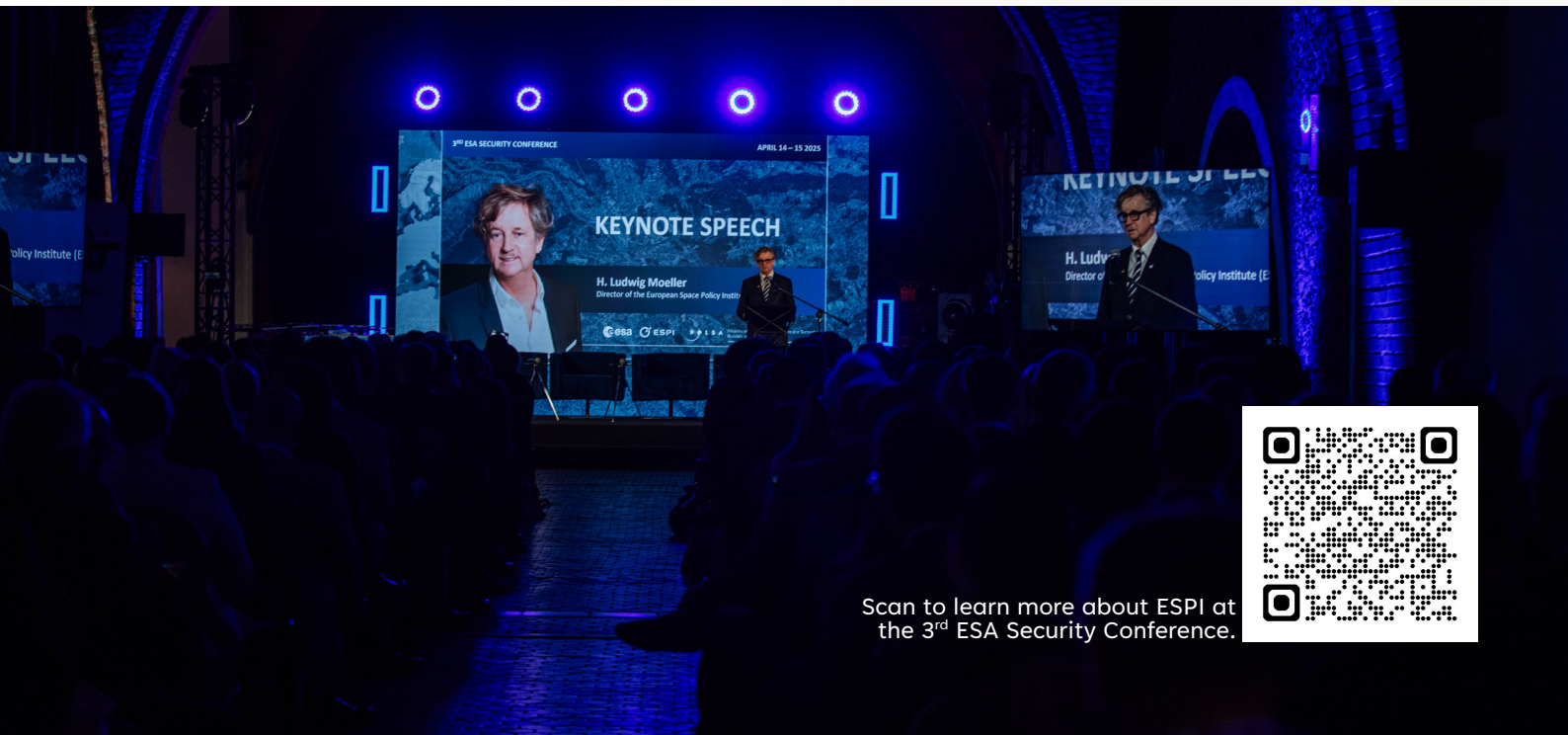
We got the village together. What is it that we need to do? We get the scale-up. We get the speed going. It was mentioned here: geo-intelligence. Maybe 30 minutes, revisit. That's a complete change from what we know. This is very demanding. Not just linear upscaling. And next to it the one thing that we need is rapid deployment. The capability that shoots up systems as they are needed. The launch sector and Earth observation—geo-intelligence—are interlinked maybe also with integrated systems: navigation, communication, together with geo-intelligence, with AI. Space systems are dependent on the launchers. And we shouldn't do any satellite program without having clear support for the launch ecosystem that is developing now.

We need, as I said multiple times now, we need to innovate. We need to demonstrate initial capabilities. This is preparatory. The development agencies can do that. We can also—and we need to—deploy operational capabilities. We all will need to work in a federation.

In summary you need top-notch R&D, innovation, and you need a system architect. And there are a few already existing in European national agencies and ESA. I would not recommend starting to discuss a European DARPA. We do not have that time now. The defence industry will help dramatically, for instance, with mass production. We need to advance in Europe—our European village—and we need to do it with international cooperation.

CONCLUSIONS

In conclusion—let us stay with our values. Let's stay together, with all of what is at hand. Let's get going with what we have. It's not perfect, but it exists. There's something on which we can build. Let's do it on a scale: 0.25% GDP for space in Europe. Let's do it at speed. Let's scale up in 2025. And the nations that have it on the agenda now—in the ministerial of ESA and the preparations in Brussels—are moving forward. And with this, I conclude, and thank you very much.



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